Humoral Immunity: Antibody Structure and Function

Lecturer:

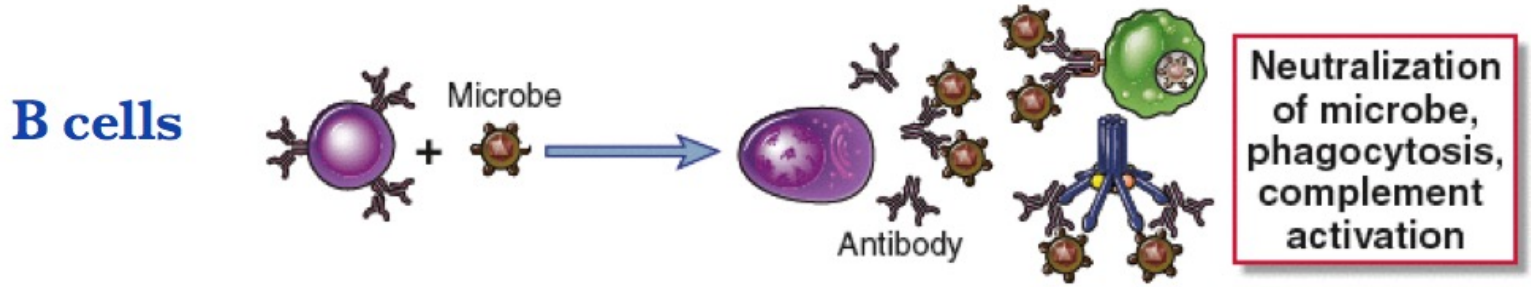
Jason Cyster

Session Objectives

- Describe the structures of the various antibody classes and how they function in the immune system.
- In general terms, explain how the processes of heavy and light chain gene segment recombination (VDJ recombination), B cell somatic hypermutation, and isotope switching all contribute to antibody production, diversity, and function.
- Explain the concept of clonal selection and how it impacts antibody production.
- Describe the principles of specificity and memory with regard to adaptive immunity.

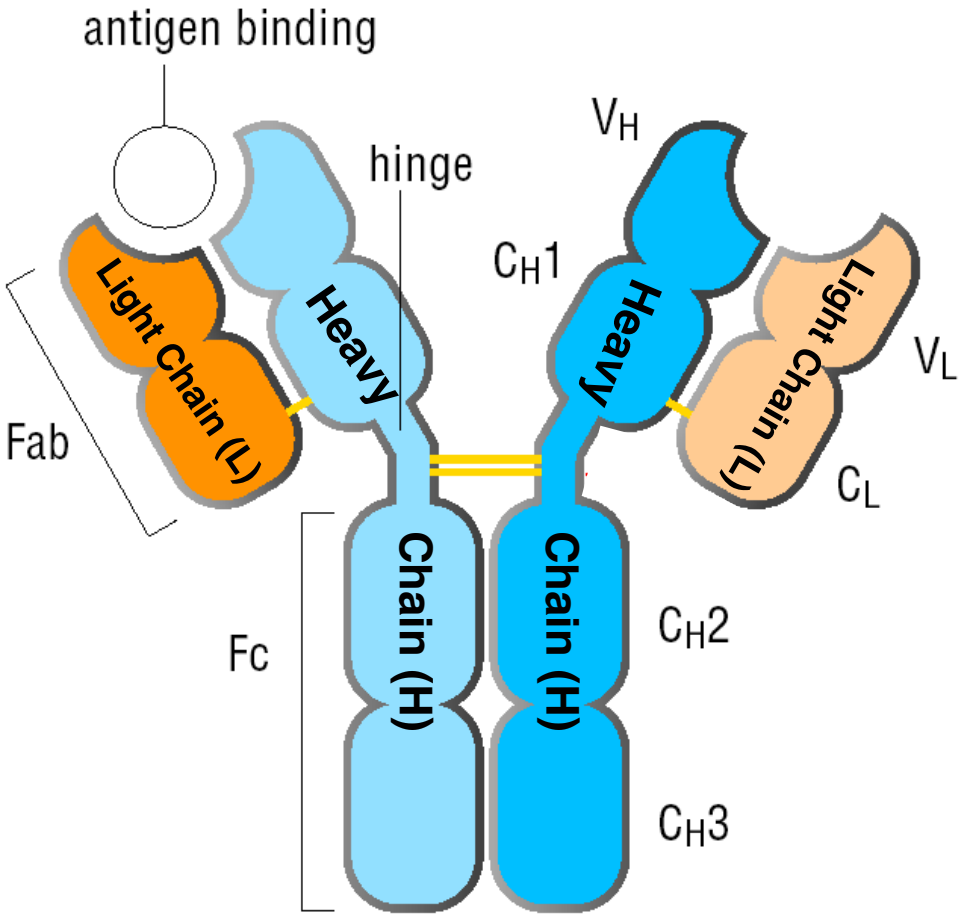
Antibodies

- Secreted proteins that protect us from pathogens
- Made by B lymphocytes (B cells)
- Bruton's agammaglobulinemia: lack B cells -> very susceptible to infection (especially bacterial)
  - ***People with congenital or acquired immune deficiencies are prone to mouth bacterial, viral and fungal infections***
- Great diversity and specificity: $>10^{10}$ possible antibodies; can distinguish between very similar molecules
- Soluble recognition; recognize proteins, polysaccharides, etc.
- Tag particles for clearance/destruction
- Protect against re-infection (vaccines)



Antibody Structure

(Immunoglobulin)



Immunoglobulin (Ig):
composed of

- Variable domains (V)
- Constant domains (C)
- Hinge

Fab: **f**ragment **a**ntigen **b**inding

Fc: **f**ragment **c**rystallizable or **c**onstant (effector functions)

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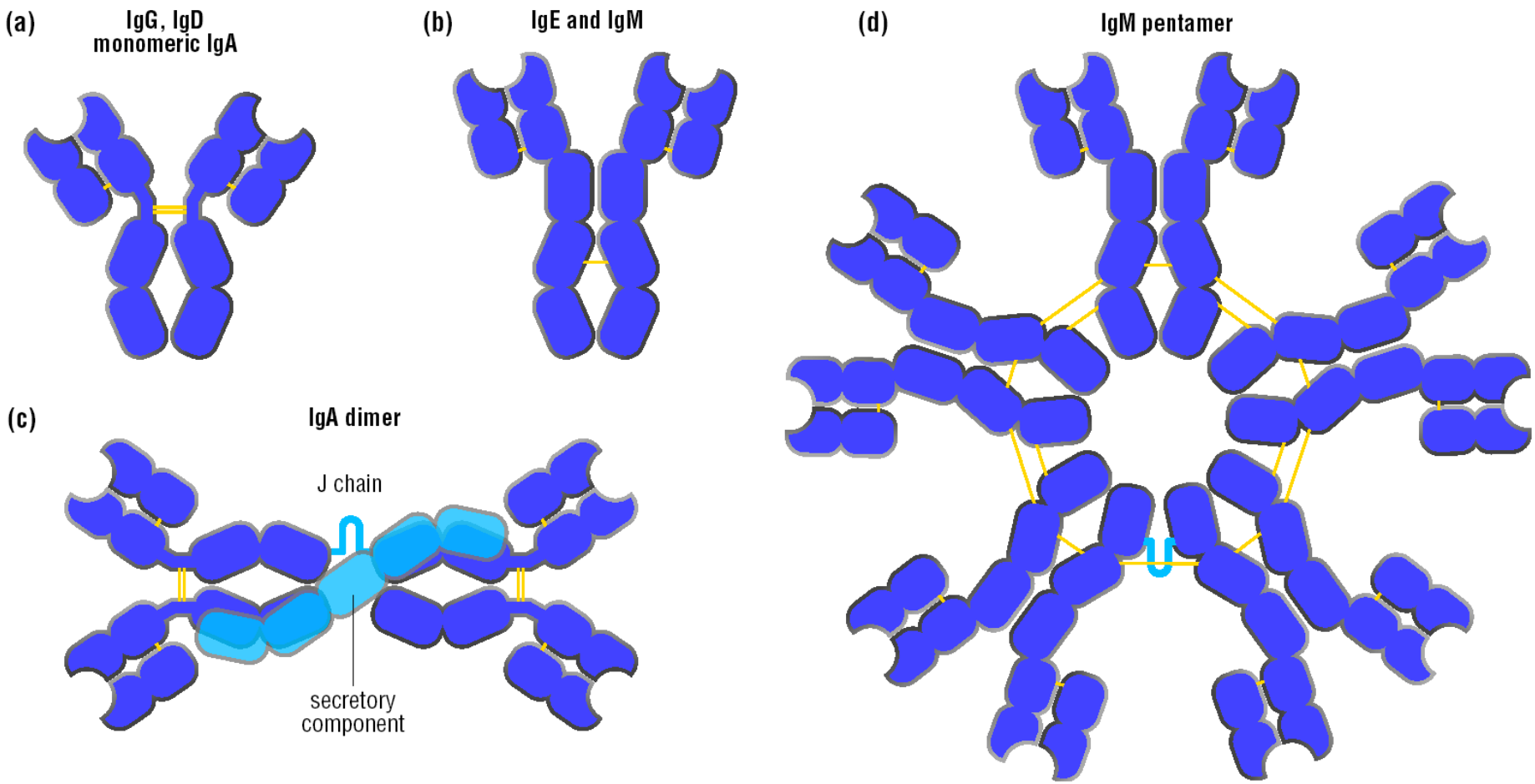
Variability in antibodies is clustered in the loops (green) in the Variable domains of the heavy and light chains; these regions are responsible for binding to antigen

5 Different Antibody Classes



- function of IgD is not well characterized and we won't be discussing further

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Antibody Isotypes (Classes) Important for Immunity

- IgM (μ)
  - First antibody produced in immune response
  - Antigen receptor (B cell receptor, BCR) of naïve B cells
- IgG (γ)
  - Major antibody of serum and extravascular tissues
  - Crosses placenta to provide protection in utero
- IgA (α)
  - Major antibody of secretions (mucus, saliva, milk, etc.)
- IgE (ε)
  - Important for immune defense against helminths; also impacts allergy and asthma

The immune system makes antibodies to antigens <u>not previously encountered</u>

(Plasticity)

How?

(1) the random generation of a very diverse array of antibodies by gene rearrangement
(2) clonal selection of B cells producing antibody specific for an antigen

Generating Antibody Diversity: VDJ Recombination

V(D)J recombination

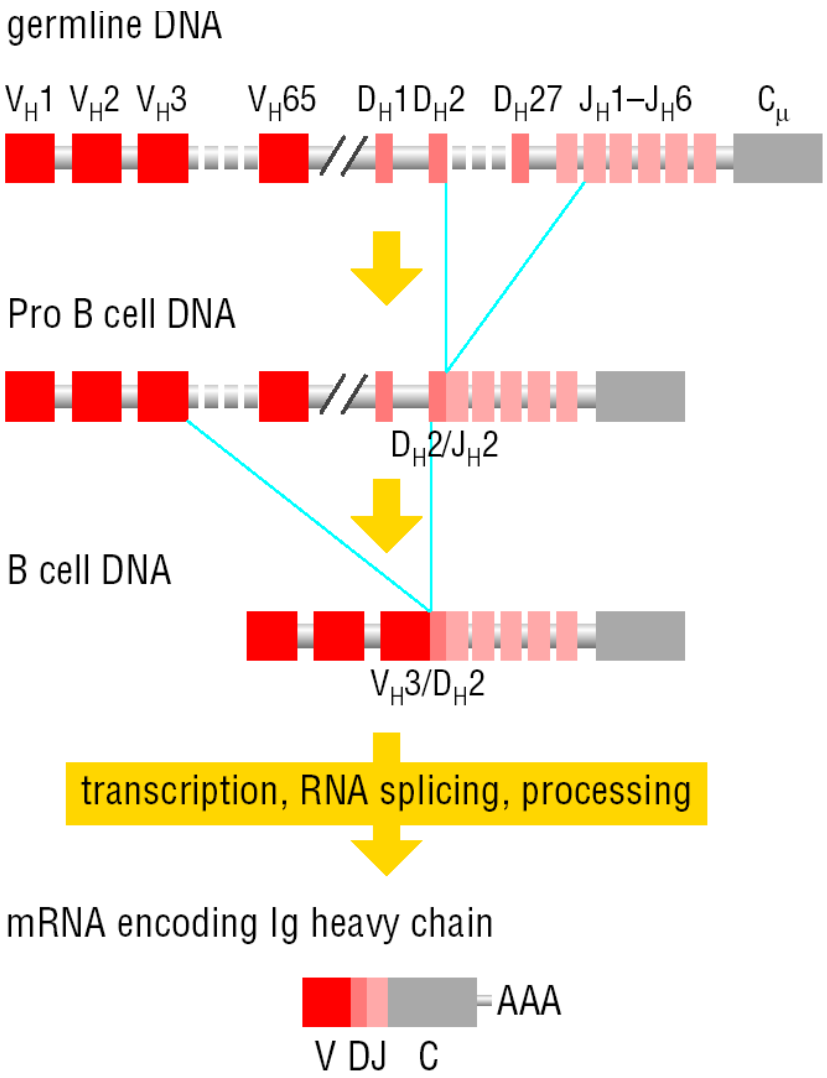
Example: Ig Heavy chain (Hc) locus

V = variable segment

D = diversity segment

J = joining segment

C = constant domain



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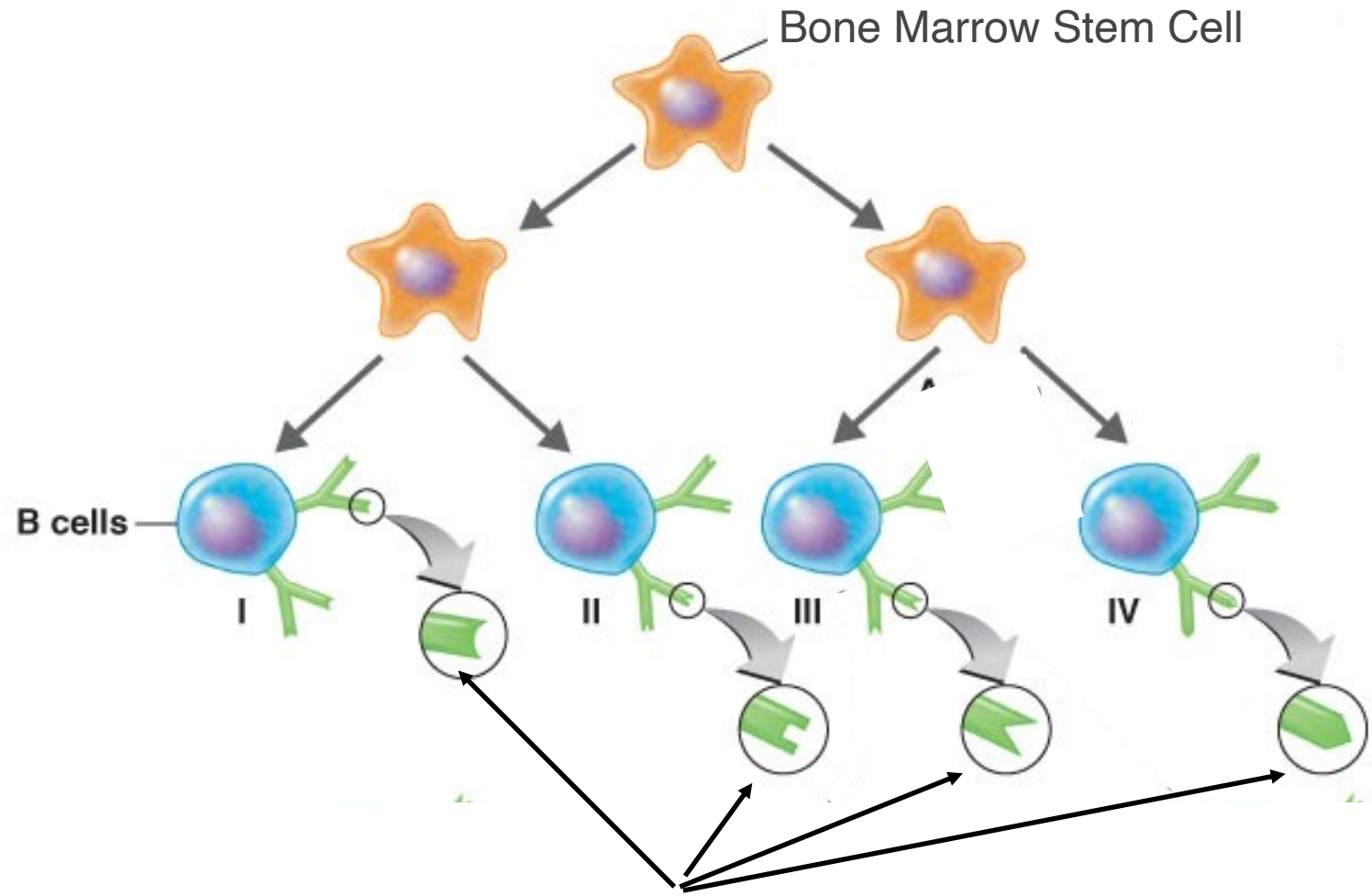
Sources of Antibody Diversity

- Heavy chains: 65 V_H x 27 D_H x 6 J_H = 10,530
- Light chains: ~40 V x ~8 J = 320
- 320 L chains x 10,530 H chains = 3.3×10^6
- Junctional diversity (addition or deletion of nucleotides at recombination sites) estimated to add 3×10^7 fold to overall diversity

(note: the exact numbers are not important)

B Cell Development

Development of BCR specificity PRECEDES exposure to antigen



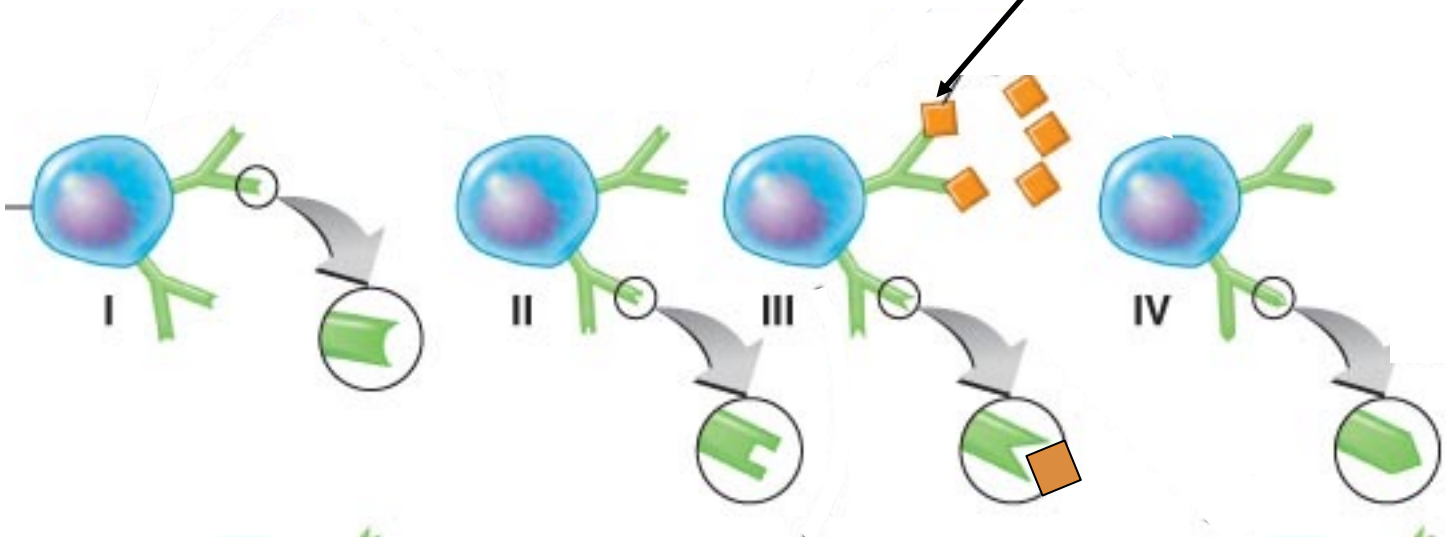
Each B cell makes a unique antigen receptor (BCR) with a single antigen-binding specificity

Antigen Recognition and Clonal Selection of Lymphocytes

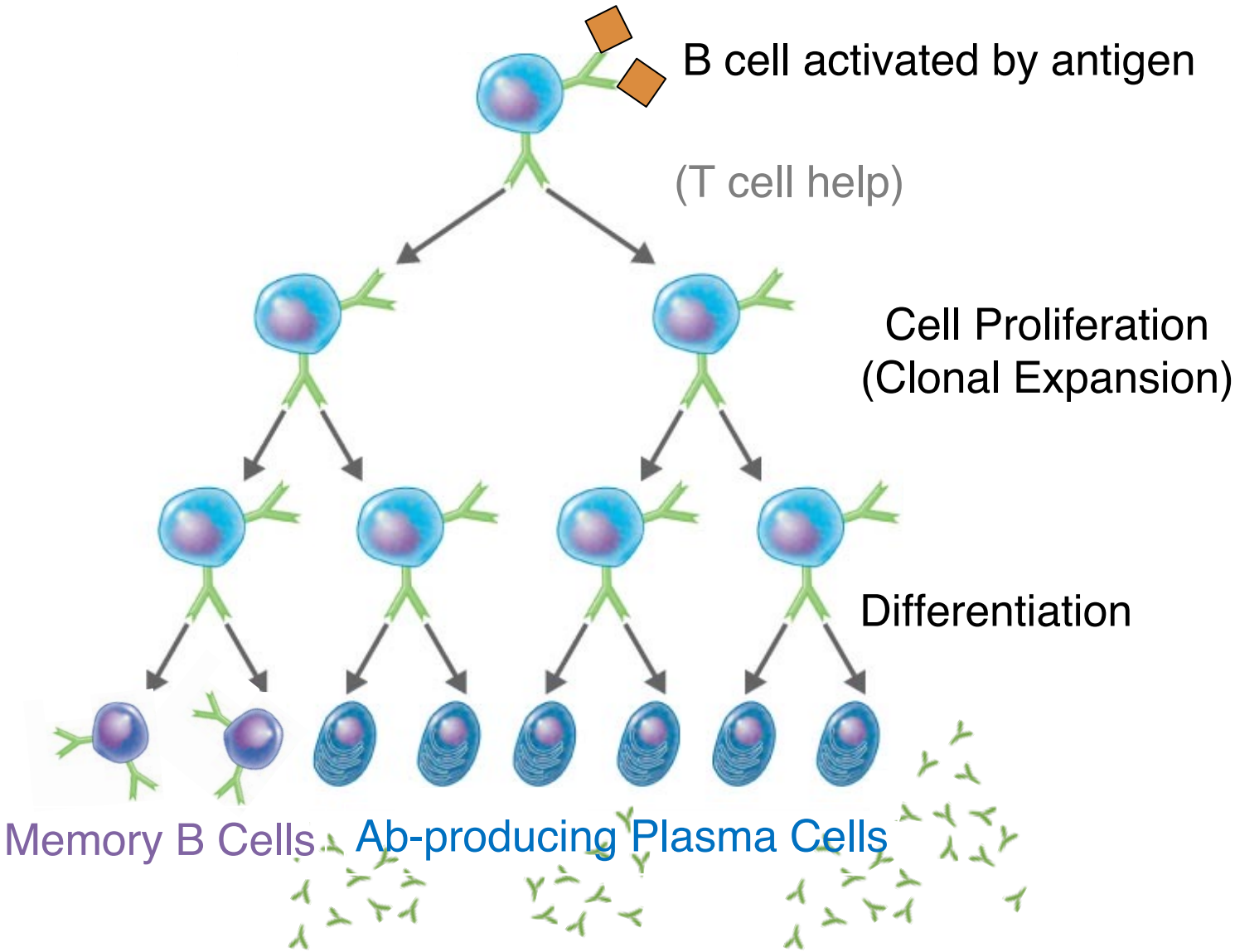
The antigen "selects" the B cell clone that **specifically** binds to it

Circulating Antigen

B cells in 2° lymph tissue

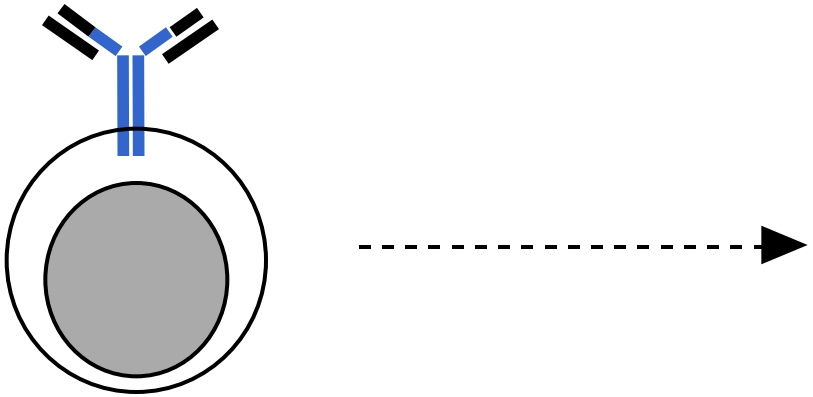


Antigen binding triggers B cell proliferation and differentiation

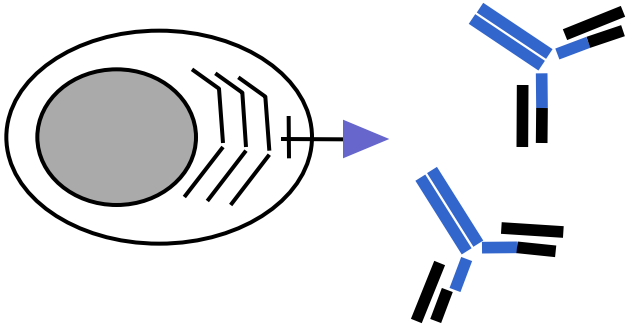


Membrane-bound vs Secreted Immunoglobulin (Ig)

membrane Ig



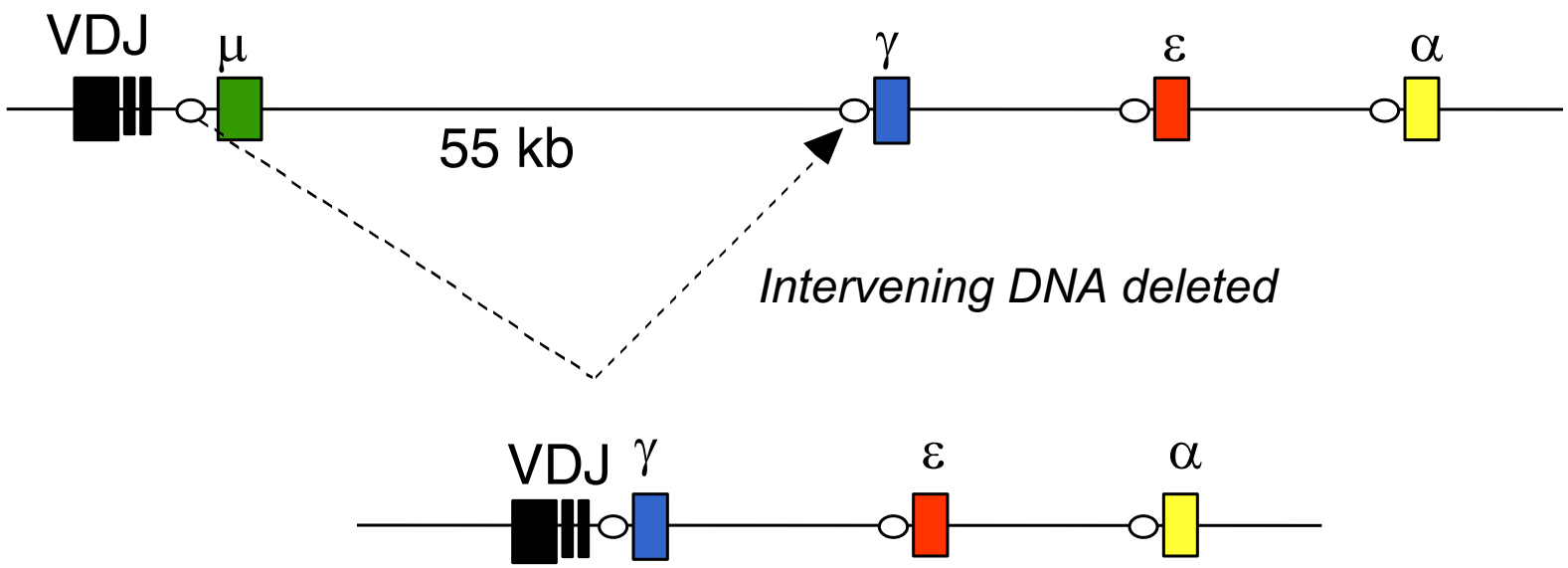
secretory Ig

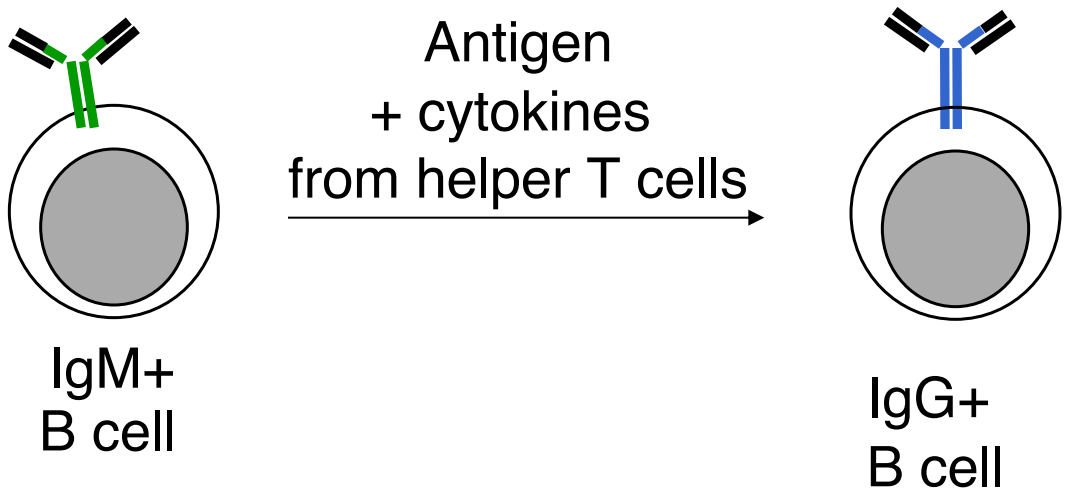


<u>B cell</u> - uses membrane Ig as surface B cell receptor (BCR) to respond to antigen and undergo clonal expansion

<u>Plasma cell</u> - makes the same antibody in a secreted form; releases ~10,000 antibody molecules per second!!

Ig Heavy chain isotype (class) switching

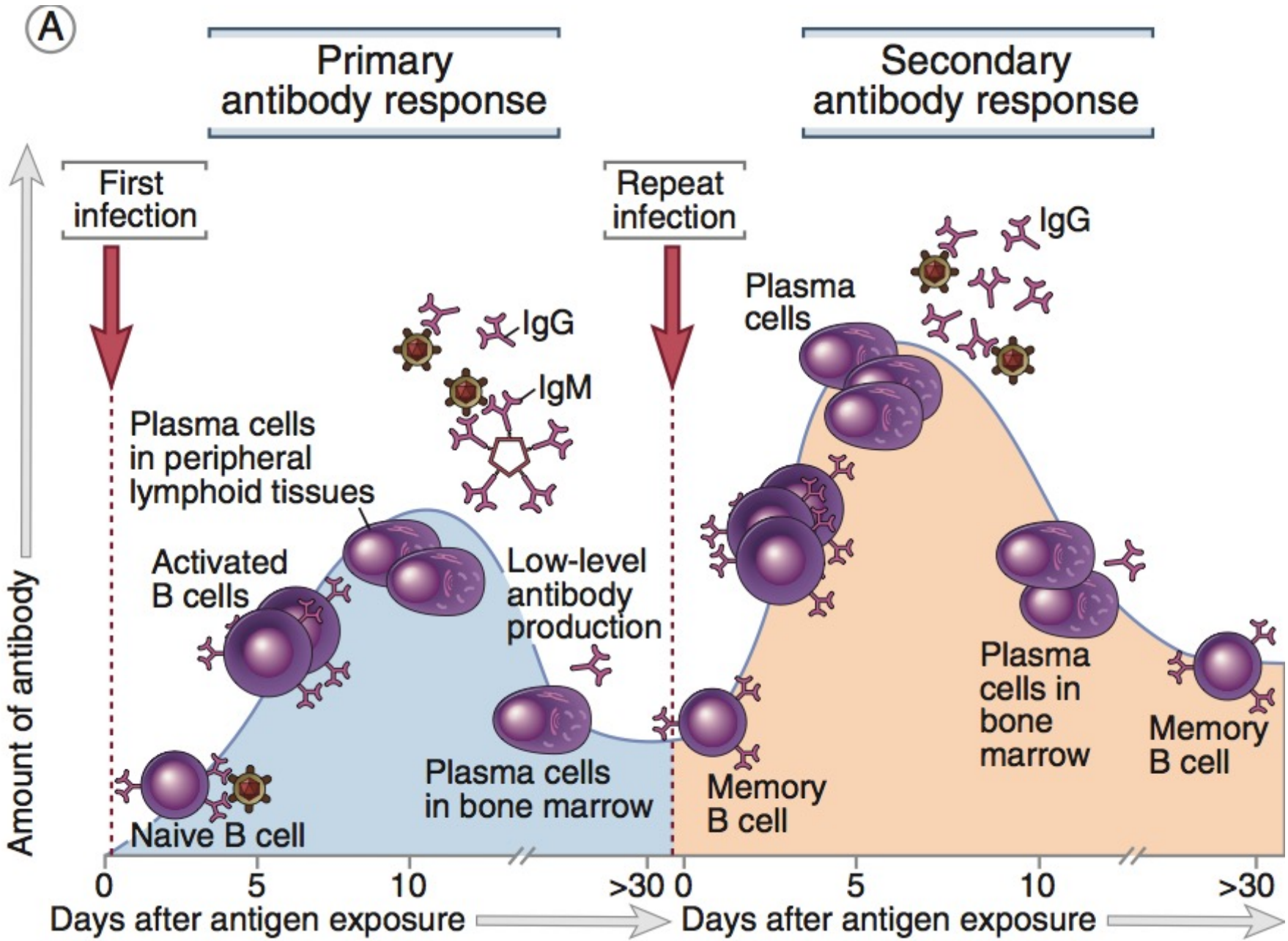




The humoral immune system has a memory

- A first encounter of a B cell with an antigen leads to secretion of antigen specific IgM (primary response)
- A second encounter with the SAME antigen leads to a more rapid and larger response dominated by IgG (secondary response)
- This improved secondary or 'memory' response is the result of changes that persist after first encounter with antigen
  - Some of the B cells that respond in the primary response do not change into plasma cells but become long-lived 'memory' B cells -> this increases the frequency of B cells specific for the antigen
- Having a higher frequency of antigen-specific B cells at the beginning of the (secondary) response causes the response to be larger and more rapid

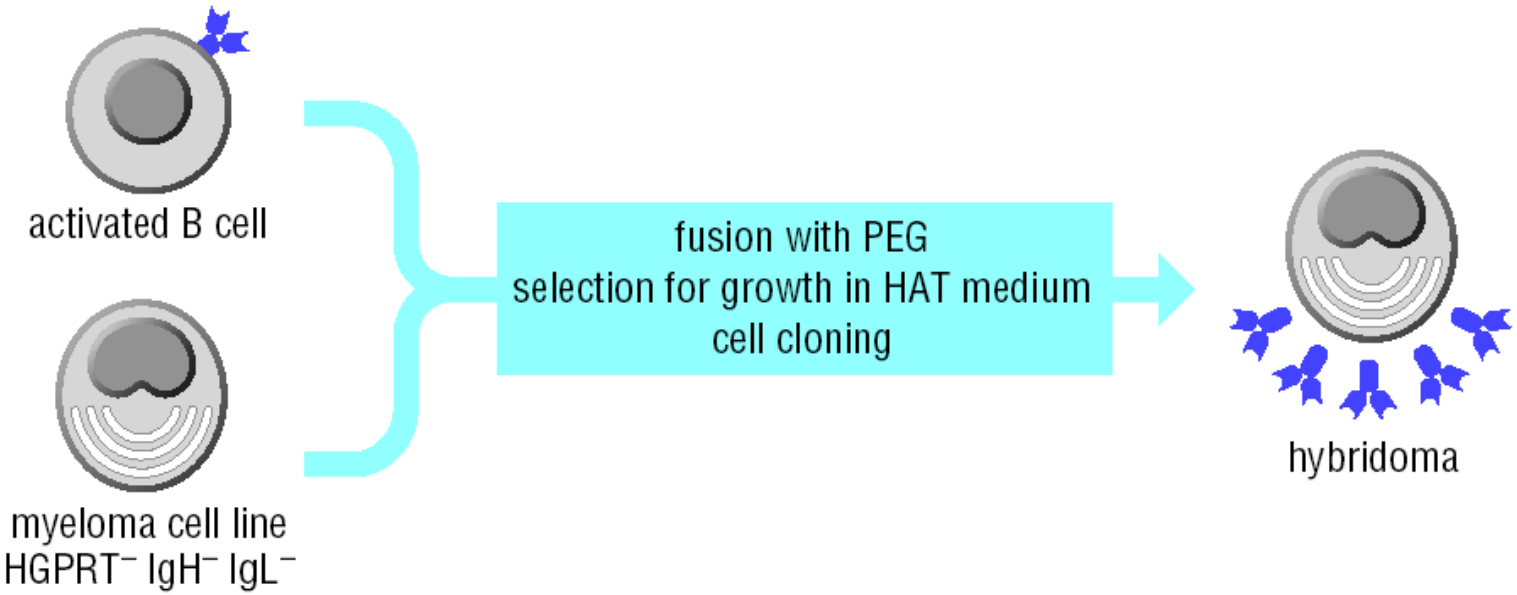
Features of primary and secondary antibody responses



Abbas Fig. 7-3

Monoclonal Antibodies

- Single antibody (all same H and L chains)
- Made by fusion of B cells to a transformed (immortal) cell line of the plasma cell type (myeloma) and selection for “hybridomas” that produce antibody with the desired properties
- Alternatively, made by cloning the Hc and Lc genes from an antigen-specific B cell and expressing the antibody in a highly secretory cell line



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Monoclonal antibodies used in medicine

Standardized, unlimited reagents for diagnosis or therapy
(human antibodies or “humanized” antibodies can be made)

Monoclonal Antibodies Used in Therapies		
monoclonal antibody	target	disease
trastuzumab	HER2	breast cancer
infliximab	TNF	rheumatoid arthritis, Crohn's disease
rituximab	CD20	non-Hodgkin's lymphoma
abciximab	GPIIb/IIIa	coronary disease
OKT3	CD3	graft rejection
Ipilimumab	CTLA4	cancer (checkpoint blockade)
Pembrolizumab	PD-1	cancer (checkpoint blockade)

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Questions

- You can make antibodies against

A) Pathogens

B) Drugs

C) Allergens

D) Almost anything foreign

E) All of the above

- Plasma cells secrete antibody of the same specificity as the precursor B cell? Yes or No

- When a B cell undergoes isotype switching from IgM to IgG do you think it keeps the same antigen specificity? Yes or No

You can make antibodies against

Pathogens

Drugs

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Almost anything foreign

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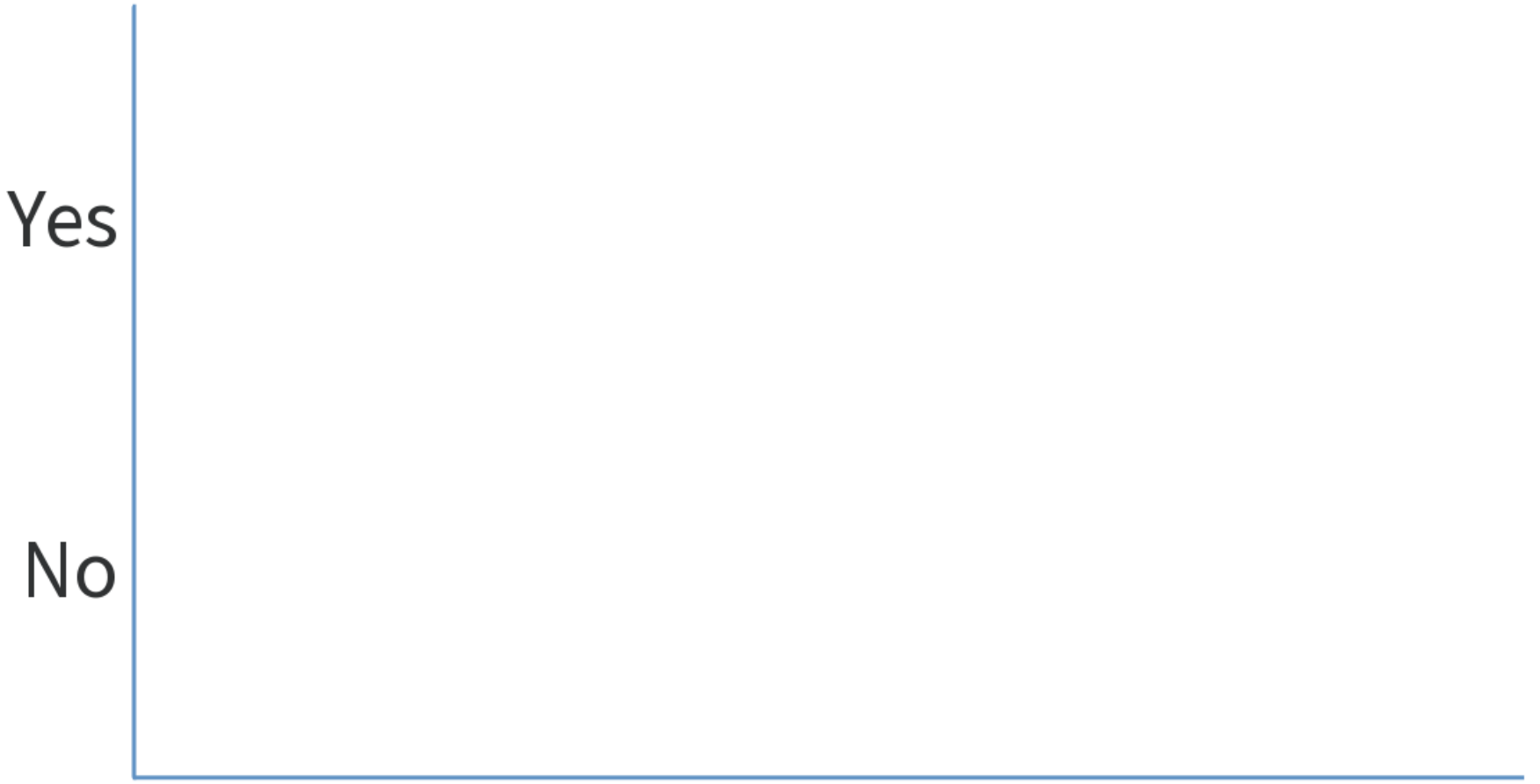
Plasma cells secrete antibody of the same specificity as the precursor B cell?



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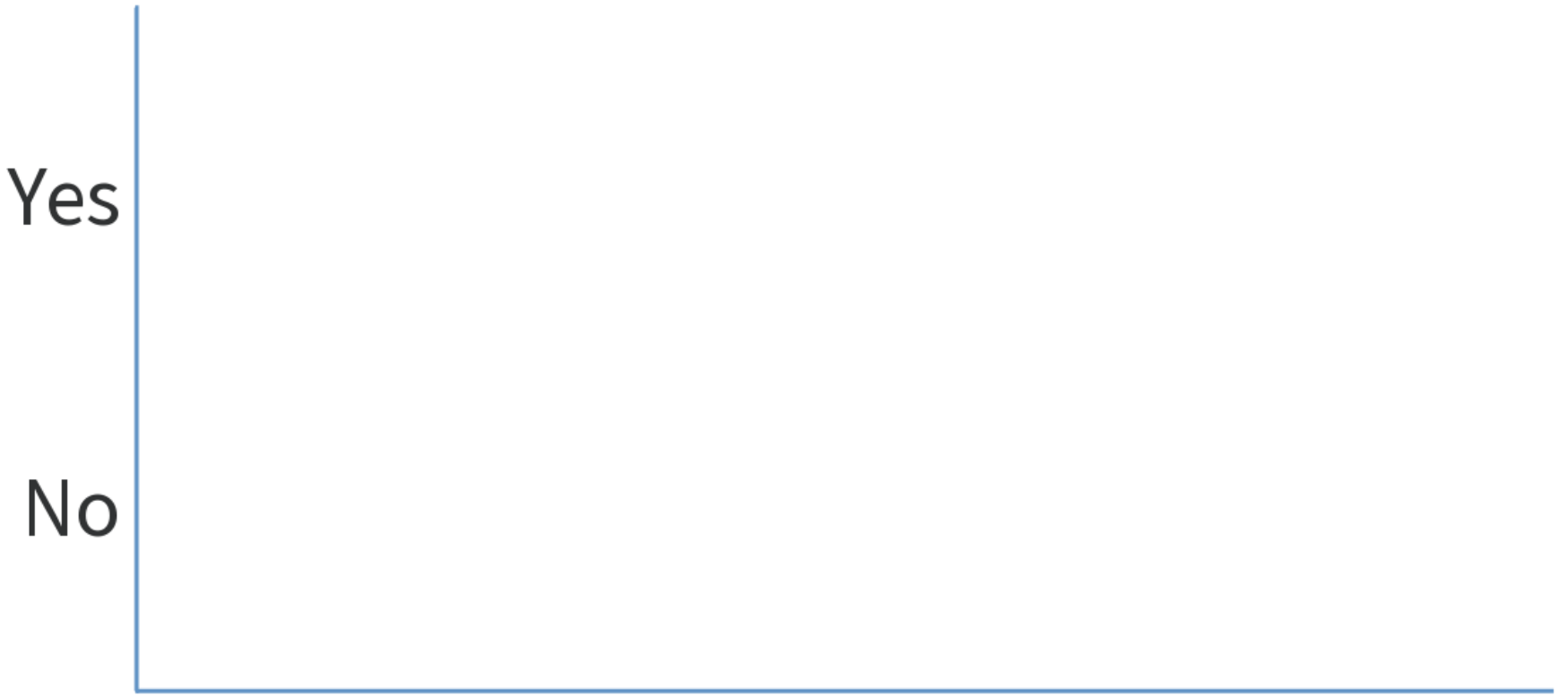
When a B cell undergoes isotype switching from IgM to IgG do you think it keeps the same antigen specificty?



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